

Lesson 7: Solving Exponential & Log Equations — Practice Worksheet

Name: _____ Date: _____

Attempt every question. Show your working. The sections increase in difficulty.

Section 1 — Warm-up (foundation)

1. Solve $2^x = 64$.
2. Solve $\log_3 x = 2$.
3. Solve $10^x = 1000$.
4. Solve $5^x = 125$.
5. Solve $\log_2 x = 4$.
6. Solve $4^x = 16$.
7. Solve $\log_{10} x = 2$.

Section 2 — Core practice

8. Solve $2^x = 20$, giving your answer to 3 s.f.
9. Solve $5^x = 90$, giving your answer to 3 s.f.
10. Solve $3^x = 40$, giving your answer to 3 s.f.
11. Solve $\log_4 x = 3$.

12. Solve $7^x = 200$ (3 s.f.).

Section 3 — Challenge & exam-style

13. Solve $3^{(x+1)} = 90$, giving your answer to 3 s.f.

14. Solve $\log_2(x) + \log_2(x - 6) = 4$.

15. Solve $2^{(2x)} = 50$ (3 s.f.).

16. Solve $\log(x) + \log(x - 3) = 1$ (log base 10).